

Palmar Dermatoglyphic Pattern in Relation to Blood Group and Gender in Saudi Population

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ABSTRACT **Background:** Forensic medicine deals with the identification of individuals, but it also involves legal documents, financial transactions, and civil affairs. One of the most established and trustworthy biometric techniques is the fingerprint, which is accepted as valid identification documentation for individuals. **Methods:** This cross-sectional research involved 420 Saudi students, including 300 males and 120 females, aged between 18 and 27. On a white A4-sized Performa, rolled imprints of each participant's ten fingers were obtained, and these were categorized into loops, whorls, arches, and composites. The relationship between the ABO and Rh blood groups and fingerprint patterns were compared using the Chi-square test to see for any correlation. **Results:** The study included 420 participants in total. Of these 420, there were males 300 (71.4%), and females 112 (28.6%) participants, respectively. Of the 420 participants, blood groups, such as A, B, AB, and O, were represented by 112 (26.7%), 75 (17.8%), 20 (4.8%), and 213 (50.7%) of the participants, respectively. Out of the 4200 fingerprints, 127 (3.0%) were composite, 1547 (36.8%) were whorl, 197 (4.7%) were arch, and 2329 (55.5%) were loop. An analysis of the fingerprint patterns of the ABO blood group revealed a statistically significant variance ($P = 0.0002$). **Conclusion:** According to this study, the most common fingerprints were loop (55.5%) and whorl (36.8%). ABO blood group differences in fingerprint patterns were found to be statistically significant.

KEYWORDS: Arch, blood groups, fingerprints, gender, loop, whorl

INTRODUCTION

Human fingerprints are thought to be the most reliable form of identification in the modern world. The pattern of fingerprints is attributed to an individual's genetic makeup and is formed during fetal development. Once formed, the pattern remains unchanged throughout a person's lifetime and is not influenced by external factors.^[1] The uniqueness and stability of fingerprints make them a valuable tool in various fields, such as identification and forensic investigations.^[2]

Sir Francis Galton, one of the most creative biologists of his day, published his classic book on fingerprints in 1892, initiating the biological study of fingerprint. It is intriguing to observe that Dr. Cummins, the pioneer of American fingerprint analysis, subsequently introduced the term "dermatoglyphics" to characterize the examination of fingerprints and their patterns. In Greek,

the word Dermatoglyphics means "skin carving," where derma is the word for skin and glyph means carving.^[3]

Previous studies on dermatoglyphics have shown a significant relationship between blood types and fingerprint patterns.^[4] According to previous studies, the main fingerprint pattern is identical for different blood groups, where loops were found to be the most common pattern, followed by whorls, and arches being the least common.^[5] Similar research done on fingerprint has found a high correlation between gender, blood group, and fingerprint pattern.^[6] Often, the only evidence remaining at a complicated site of the crime that can

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help to recognize the suspect or the criminals is the fingerprints and some blood stains.^[7]

In spite of the reality that human beings have been using fingerprints as a means of identification for quite a period of time, but the goal of this study is to figure out that if there is any evidence of a significant relationship between blood group, fingerprint patterns, and gender.

MATERIAL AND METHOD

Study design and population

The participants in this observational cross-sectional study were both male and female students enrolled in Majmaah University's Health Sciences College in the Kingdom of Saudi Arabia. The sample population's age ranged from 18 to 27 years old.

Inclusion Criteria:- Healthy people, or those who were at least 18 years old and above and free of any conditions that might alter fingerprints, were included.

Exclusion Criteria:- People with blood group disorders, ink pad allergies, hand or finger deformities, or missing fingers were not included.

All the selected individuals were called upon to take part in the study and a consent in writing was acquired from the participant. The study participants were enrolled using a universal sampling technique and consent was taken before procedure.

Study procedure

Prints were made on the white proforma in the A4 size. Upon gathering the sheets bearing the prints of both hands' fingers, the fingerprints were examined and categorized with the aid of a magnifying glass. The subjects' blood groups were taken from their driver's license, which was based on legitimate medical records to prevent any false information from tainting the results.

The fingerprint pattern types were categorized in accordance with Sandhu *et al.*'s^[8] previous description. Each finger's fingerprint pattern was recorded, along with the details of the participants.

Statistical analysis

IBM SPSS 25.0 was used to analyze the data after it had been collected and entered into Excel spreadsheets. The one-sample Kolmogorov–Smirnov test was utilized to assess the statistical significance of the results. A *P* value below 0.05 suggested that the value could be deemed statistically significant.

RESULT

In our research, 71.4% of the participants were males, resulting in a male-to-female ratio of 2.5:1. The most

prevalent blood group among the participants was O (50.7%), followed by A (26.7%), with AB (4.8%) being the least common. The Rh factor classification showed that O+ve (47.1%) was the most common blood group among the participants, and AB-ve (0.7%) was the least common.

Loop in males (54.6%) and females (57.6%) had the most common fingerprint pattern, followed by whorls males (37.5%) and females (34.4%), and the least common was composite males (3.1%) and females (2.8%) [Figure 1]. The fingerprint patterns of males and females did not differ statistically variation, as Table 1 demonstrates (*P* = 0.357).

Figure 1 showcases representative fingerprint patterns from the research subject as examples:-

The distribution of fingerprint patterns by ABO group is displayed in Table 2. The blood group O had the maximum number of loop, followed by A, B, and AB. The blood group O had the highest whorl, followed by A, B, and AB. It was discovered that there was a statistically significant difference in fingerprint patterns amongst the ABO blood group (*P* = 0.0002).

The distribution of fingerprint patterns in each of the four ABO blood groups' Rh+ve and Rh-ve individuals is shown in Table 3. The most prevalent blood groups in Arches are O+ve (39%) and A-ve (5%), and O-ve (5%). However, it was discovered that blood groups A+ve and O-ve were significant.

DISCUSSION

The morphological features of fingerprints are not only specific to each person but also do not alter over the

Table 1: Distribution of fingerprint in all fingers of both hands by gender (n=4200)

Fingerprint pattern	Gender						Chi-square test, <i>P</i>
	Male		Female		Total		
	<i>n</i>	%	<i>n</i>	%	<i>n</i>	%	
Loop	1638	54.6	691	57.6	2329	55.5	17.086, 0.357
Whorl	1134	37.8	413	34.4	1547	36.8	
Arches	135	4.5	62	5.2	197	4.7	
Composite	93	3.1	34	2.8	127	3.0	
Total	3000	100	1200	100	4200	100	

Table 2: Distribution of primary fingerprints patterns in all fingers by ABO blood group (n=4200)

Fingerprint pattern	Blood groups				Chi-square test, <i>P</i>
	A	B	AB	O	
Loop	678	432	112	1107	29.84, 0.0002
Whorl	362	270	58	857	
Arches	51	35	24	87	
Composite	29	13	6	79	
Total	1120	750	200	2130	



Figure 1: Types of fingerprint pattern

Table 3: Distribution of fingerprint in all fingers of both hands with regard to ABO rhesus blood group (n=4200)

Fingerprint pattern	Loop (n=2329)	Whorl (n=1547)	Arch (n=197)	Composite (n=127)	Total	Chi-square test	P
Blood group							
A +ve	645 (27.7%)	331 (21.4%)	41 (20.8%)	23 (18.1%)	1040	21.53	<0.001
A -ve	33 (1.4%)	31 (2.0%)	10 (5.0%)	6 (4.7%)	80		
B +ve	376 (16.2%)	258 (16.7%)	35 (17.8%)	11 (8.7%)	680	34.73	0.762
B -ve	56 (2.4%)	12 (0.8%)	0 (0%)	2 (1.6%)	70		
AB +ve	96 (4.1%)	48 (3.1%)	20 (10.2%)	6 (4.7%)	170	29.65	0.548
AB -ve	16 (0.7%)	10 (0.6%)	4 (2.0%)	0 (0%)	30		
O +ve	1055 (45.3%)	753 (48.7%)	77 (39.1%)	75 (59.0%)	1960	24.95	0.001
O -ve	52 (2.2%)	104 (6.7%)	10 (5.0%)	4 (3.2%)	170		
Total	2329 (100%)	1547 (100%)	197 (100%)	127 (100%)	4200		

course of a lifetime.^[9] Male participants constituted 71.4% of the study, while female participants made up 28.6%. The respondents' most common blood group was O (50.7%), which was followed by A, B, and, less frequently AB blood group. Studies by Bharadwaja *et al.*^[4] from India, Fayrouz *et al.*^[10] from Libya, Al-Riyami *et al.*^[11] from Oman, and Kc *et al.*^[12] from Nepal reported similar results.

Nevertheless, in research conducted by Khalid and Qureshi^[13] in Pakistan, Ghasemi *et al.*^[14] in Iran, and Mehta and Sonar^[15] in India, they observed that blood group "B" emerged as the most prevalent, and blood group "AB" had the lowest proportion, while Jaff and O'Briain discovered that blood groups "O" (43%) and "A" (40%), "B" (12%), and "AB" (7%) were more common in Europeans.^[16]

Overall, loops males (54.6%) and females (55.5%) were the most prevalent fingerprint pattern found in our research, followed by whorls males (37.8%), a female (36.8%), and Arch males (4.5%) and females (4.7%). Most of the previous studies by Fayrouz *et al.*^[10] and Smail *et al.*^[17] showed similar results. However, whorl was found to be the most prevalent pattern in an Indian population study by Thakur *et al.*^[18]

Upon analyzing the distribution of fingerprint patterns within the ABO-Rh blood groups, we identified that

the loop frequency was higher in the O+ve (45.3%), A+ve (27.7%), B+ve (16.2%), groups of individuals. However, whorl was greater in the groups A+ve (21.4%) and O+ve (48.7%). Research conducted in Libya by Fayrouz *et al.*^[10] in India by Mehta and Sonar,^[15] and in Nepal by Kc *et al.*^[12] demonstrated a higher prevalence of the loop.

Previous research (Fayrouz *et al.*, 2012; Smail *et al.*, 2019; Kc *et al.*) also revealed that the arch pattern had the lowest incidence among the primary fingerprint patterns in all blood groups.^[10,12,17] Nevertheless, as reported by Mehta and Sonar,^[15] the arch pattern prevailed as the most common in the A-negative, B-positive, and O-negative blood groups. In our investigation, comparable findings were observed only in the A-positive and O-positive blood groups.

CONCLUSION

The loop pattern is the most frequently observed fingerprint pattern in both males and females, followed by whorls and composites, which are less common fingerprint patterns. An individual's fingerprint pattern can be used to predict their ABO blood group and vice versa, improving the accuracy of determining whether an individual is alive or deceased. It will be exciting to

see how this technology develops in the coming years which could enhance the accuracy and reliability of identification.

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Conflicts of interest

There are no conflicts of interest.

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